$$= 9\left(\frac{t-2}{2t+1}\right)^8\left[\frac{(2t+1)\frac{d}{dt}(t-2)-(t-2)\frac{d}{dt}(2t+1)}{(2t+1)^2}\right]$$

$$= 9\left(\frac{t-2}{2t+1}\right)^8\left[\frac{(2t+1)\times 1-(t-2)\times 2}{(2t+1)^2}\right]$$

$$= 9\left(\frac{t-2}{2t+1}\right)^8\left[\frac{2t+1-2t+4}{(2t+1)^2}\right]$$

$$= \frac{45(t-2)^8}{(2t+1)^{10}}\ .$$

Example 10.13

Differentiate $(2x + 1)^5\ (x^3 - x + 1)^4$.

Solution

Let $y = (2x + 1)^5\ (x^3 - x + 1)^4$

Take $u = 2x + 1$; $v = x^3 - x + 1$ so that

$y = u^5 \cdot v^4$

$$\frac{dy}{dx} = u^5.\frac{d}{dx}(v^4)+v^4\frac{d}{dx}(u^5)$$ by Product Rule

$$= u^5.4v^{4-1}\frac{dv}{dx}+v^4.5u^{5-1}\frac{du}{dx}$$ by Chain Rule

$$= 4u^5.v^3\times(3x^2-1)+5v^4u^4\times 2$$

$$= 4(2x+1)^5(x^3-x+1)^3(3x^2-1)+10(x^3-x+1)^4(2x+1)^4$$

$$= (2x+1)^4(x^3-x+1)^3\left[4(2x+1)(3x^2-1)+10(x^3-x+1)\right]$$

$$= 2(2x+1)^4(x^3-x+1)^3(17x^3+6x^2-9x+3)\,.$$

Example 10.14

Differentiate : $y = e^{\sin x}$.

Solution

Take $u = \sin x$ so that

$y = e^u$

$$\frac{dy}{dx} = \frac{d(e^u)}{du}\times\frac{du}{dx} = e^u \times \cos x = \cos x\, e^{\sin x}.$$

Example 10.15

Differentiate 2^x.

Solution

Let $y = 2^x = e^{x\log 2}$.

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